

KAIYUAN TANG

CONTACT INFORMATION

356B Fitzpatrick Hall
University of Notre Dame, Notre Dame, IN
Phone: +1 409-791-2729
Email: ktang2@nd.edu
Website: www.kaiyuan-tang.com

RESEARCH INTERESTS

Machine Learning for Scientific Data (Neural Representations, Compression, Generalization)
Scientific Visualization (Feature Analysis, Interactive Exploration, Visual Communication)
Computer Graphics (Expressive Rendering, Editable Scene Representations)
Human-Centered AI (Multimodal Interaction, Visualization Agents, Benchmarking and Evaluation)

EDUCATION

Ph.D., Computer Science & Engineering, Aug 2022–Jun 2026
University of Notre Dame, Notre Dame, IN
Advisor: Prof. Chaoli Wang

Bachelor of Engineering, Telecommunications Engineering, Sep 2018–Jun 2022
Xidian University, Xi'an, Shaanxi, China

EMPLOYMENT

Postdoctoral Researcher, Aug 2026–Present
University of Notre Dame, Notre Dame, IN
Sponsor: Prof. Chaoli Wang

AWARDS & RECOGNITIONS

CSE Outstanding Research Assistant Award, 2026
IEEE VIS Travel Grant, 2024
Notebaert Professional Development Fund, 2024
Third-class Scholarship, Xidian University, 2020–2022
Finalist Prize, The Mathematical Contest in Modeling, 2021

PUBLICATIONS

Eight first- or co-first-authored full papers published or accepted at leading venues, including IEEE TVCG/VIS, CVPR, IEEE PacificVis, and Computers & Graphics. [[Google Scholar](#)]
Letters for numbering: J — Journal Article; C — Conference Paper; S — Submitted.

Refereed Journal Articles

- [TVCG] [J10] **Kaiyuan Tang**, Maizhe Yang, and Chaoli Wang. EVOLVE: Efficient Learned Volume Compression with Variable-Rate Encoding on a Cross-Domain Database. *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2026)*, 33(1), Jan 2027. [[URL](#)] [145/591=24% Acceptance Rate] [**Presented at IEEE VIS 2026**]
- [TVCG] [J9] **Kaiyuan Tang** and Chaoli Wang. ECoNGS: Efficient Compressive Neural Gaussian Splats for Volume Visualization. *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2026)*, 33(1), Jan 2027. [[URL](#)] [145/591=24% Acceptance Rate] [**Presented at IEEE VIS 2026**]
- [TVCG] [J8] Kuangshi Ai, Haichao Miao, **Kaiyuan Tang**, Nathaniel Gorski, Jianxin Sun, Guoxi Liu, Helgi I. Ingólfsson, David Lenz, Hanqi Guo, Hongfeng Yu, Teja Leburu, Michael Molash, Bei Wang, Tom Peterka, Chaoli Wang, and Shusen Liu. SciVisAgentBench: A Benchmark for Evaluating Scientific Data Analysis and Visualization Agents. *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2026)*, 33(1), Jan 2027. [[URL](#)] [145/591=24% Acceptance Rate] [**Presented at IEEE VIS 2026**]

- [TVCG] [J7] Suemin Jeon, **Kaiyuan Tang**, Chaoli Wang, and Won-Ki Jeong. Super-Gaussian: Interactive Scene Editing for 3D Gaussian Splatting and NLI-Based Volume Visualization in Virtual Reality. *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2026)*, 33(1), Jan 2027. [URL] [145/591=24% Acceptance Rate] [Presented at IEEE VIS 2026]
- [TVCG] [J6] **Kaiyuan Tang**, Kuangshi Ai, Jun Han, and Chaoli Wang. TexGS-VolVis: Expressive Scene Editing for Volume Visualization via Textured Gaussian Splatting. *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2025)*, 32(1):933-943, Jan 2026. [URL] [131/537=24% Acceptance Rate] [Presented at IEEE VIS 2025]
- [TVCG] [J5] Jun Han, **Kaiyuan Tang**, and Chaoli Wang. MoE-INR: Implicit Neural Representation with Mixture-of-Experts for Time-Varying Volumetric Data Compression. *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2025)*, 32(1):254-264, Jan 2026. [URL] [131/537=24% Acceptance Rate] [Presented at IEEE VIS 2025]
- [TVCG] [J4] Kuangshi Ai, **Kaiyuan Tang**, and Chaoli Wang. NLI4VolVis: Natural Language Interaction for Volume Visualization via LLM Multi-Agents and Editable 3D Gaussian Splatting. *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2025)*, 32(1):46-56, Jan 2026. [URL] [131/537=24% Acceptance Rate] [Best Paper Award] [Presented at IEEE VIS 2025]
- [TVCG] [J3] **Kaiyuan Tang**, Siyuan Yao, and Chaoli Wang. iVR-GS: Inverse Volume Rendering for Explorable Visualization via Editable 3D Gaussian Splatting. *IEEE Transactions on Visualization and Computer Graphics (IEEE PacificVis 2025)*, 31(6):3783-3795, Jun 2025. [URL] [14/123=11% Journal Track Acceptance Rate] [Presented at IEEE PacificVis 2025]
- [TVCG] [J2] **Kaiyuan Tang** and Chaoli Wang. StyleRF-VolVis: Style Transfer of Neural Radiance Fields for Expressive Volume Visualization. *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2024)*, 31(1):613-623, Jan 2025. [URL] [124/557=22% Acceptance Rate] [Presented at IEEE VIS 2024]
- [CAG] [J1] **Kaiyuan Tang** and Chaoli Wang. STSR-INR: Spatiotemporal Super-Resolution for Multivariate Time-Varying Volumetric Data via Implicit Neural Representation. *Computers & Graphics*, 119:103874, Apr 2024. [URL]

Refereed Conference Papers

- [VISxGENAI] [C6] Haichao Miao, Zhimin Li, Kuangshi Ai, **Kaiyuan Tang**, Chaoli Wang, Peer-Timo Bremer, and Shusen Liu. Toward AI VIS Co-Scientists: An Agent Harness for End-to-End Design of Scientific Visualization Applications. *2nd Workshop on GenAI, Agents, and the Future of VIS (VISxGenAI at IEEE VIS)*, Boston, MA, Nov 2026. [Workshop] [Accepted with Oral Presentation]
- [VIS SP] [C5] **Kaiyuan Tang**, Daniel Burke, and Chaoli Wang. Lossless-INR: Lossless Volumetric Implicit Neural Representations. In *Proceedings of IEEE VIS Conference (Short Papers)*, Boston, MA, Nov 2026. [URL] [68/207=32% Acceptance Rate]
- [CSCW DEMO] [C4] Niloofar Sayadi, **Kaiyuan Tang**, Yunhao Xing, Simret Gebreegziabher, Chaoli Wang, and Diego Gómez-Zarà. CoBranchMR: Supporting Parallel Design and Conflict Resolution in Mixed Reality. In *Proceedings of ACM Conference on Computer-Supported Cooperative Work and Social Computing (Demonstrations)*, Salt Lake City, UT, Oct 2026. [URL] [9/22=40% Acceptance Rate]
- [CVPR] [C3] Zixuan Pan*, **Kaiyuan Tang***, Jun Xia, Yifan Qin, Lin Gu, Chaoli Wang, Jianxu Chen, and Yiyu Shi. SGI: Structured 2D Gaussians for Efficient and Compact Large Image Representation. In *Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Denver, CO, pages 12162-12172, Jun 2026. [URL] [4,071/16,092=25% Acceptance Rate] [Highlight Presentation] (* equal contribution)
- [PACIFICVIS VN] [C2] Maizhe Yang, **Kaiyuan Tang**, and Chaoli Wang. Meta-INR: Efficient Encoding of Volumetric Data via Meta-Learning Implicit Neural Representation. In *Proceedings of IEEE Pacific Visualization Conference (Visualization Notes)*, Taipei, Taiwan, pages 246-251, Apr 2025. [URL] [12/28=42% Acceptance Rate]

- [PACIFICVIS] [C1] **Kaiyuan Tang** and Chaoli Wang. ECNR: Efficient Compressive Neural Representation of Time-Varying Volumetric Datasets. In *Proceedings of IEEE Pacific Visualization Conference*, Tokyo, Japan, pages 72-81, Apr 2024. [URL] [27/90=30% Acceptance Rate]

Manuscripts Under Review

- [CHI] [S5] Niloofar Sayadi, Simret Araya Gebreegziabher, **Kaiyuan Tang**, Yunhao Xing, Chaoli Wang, and Diego Gómez-Zarà. CoBranchMR: Supporting Parallel Design and Conflict Resolution in Mixed Reality. Submitted to *ACM CHI Conference on Human Factors in Computing Systems (CHI 2027)*.
- [TVCG] [S4] Patrick Phuoc Do, **Kaiyuan Tang**, Kuangshi Ai, and Chaoli Wang. SVLAT: Scientific Visualization Literacy Assessment Test. Submitted to *IEEE Pacific Visualization Conference 2027 (TVCG journal track)*.
- [TVCG] [S3] Guanqun Ma, David Lenz, **Kaiyuan Tang**, Hanqi Guo, Chaoli Wang, Tom Peterka, and Bei Wang. Direct Topology Tracking in Continuous Implicit Models. Submitted to *IEEE Pacific Visualization Conference 2027 (TVCG journal track)*.
- [TVCG] [S2] **Kaiyuan Tang** and Chaoli Wang. MAGIC: Multi-Level Rate-Aware Grid-Based Implicit Neural Compression for Volume Visualization. Submitted to *IEEE Pacific Visualization Conference 2027 (TVCG journal track)*.
- [SciVis CONTEST] [S1] Haichao Miao, Zhimin Li, Kuangshi Ai, **Kaiyuan Tang**, Chaoli Wang, Peer-Timo Bremer, and Shusen Liu. Multi-Agent Harness for Autonomous Multi-linked Views Visual Analytic Systems Design: An Exploratory Study on the IEEE SciVis Contest 2026. Submitted to *IEEE VIS 2026 SciVis Contest*.

INVITED TALKS Data Visualization
NSF RET Summer Program for High School Teachers, University of Notre Dame, Jun 24, 2024

EXTERNAL SERVICES **Program Committee**
IEEE PacificVis Papers Conference Track, 2027
IEEE VIS Uncertainty Visualization Workshop, 2026

Student Program Committee

Eurographics Symposium on Parallel Graphics and Visualization, 2026

Selected Journal Reviewer

Computers & Graphics
Computer Graphics Forum
Pattern Recognition
Engineering Applications of Artificial Intelligence

Selected Conference Reviewer

IEEE VIS Full Paper, 2025–2026
IEEE PacificVis TVCG Journal Track, 2025–2027
IEEE PacificVis Conference Track, 2025–2027
EuroVis Full Paper, 2026

Student Volunteer

IEEE VIS Conference, 2024

LAST UPDATED October 1, 2026